Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Tutorial - 11 **Date: Nov 9, 2023**

1. For the network shown in Figure 1, let $V_{CC} = 15V,\ \beta = 160,\ \ R_{F1} = 120k\Omega,\ \ R_{F2} = 68k\Omega, R_C = 3k\Omega,$ $R_E = 1k\Omega$. Compute

   (a) r_e

   (b) Z_i

   (c) Z_o

   (d) A_v

   (e) A_i

   Assume the reactance of the capacitors to be very small for the frequency of operation.

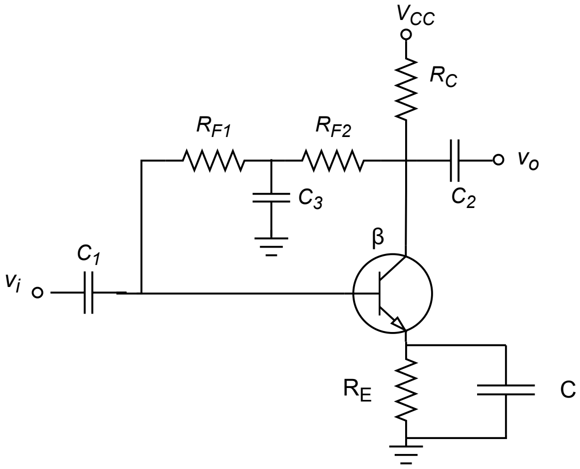


Figure 1

Solution: (a) DC Analysis:

With the capacitors open

$$
\begin{aligned}
I_B &= \frac{V_{CC} - V_{BE}}{R_F + (\beta + 1)(R_E + R_C)} \\
&= \frac{15V - 0.7V}{(120k\Omega + 68k\Omega) + (161)(3k\Omega + 1k\Omega)} \\
&= \frac{14.3V}{832k\Omega} \\
&= 17.18\mu A
\end{aligned}
$$

$$
\begin{aligned}
I_E &= (\beta + 1)I_B \\
&= (161)(17.18\mu A) \\
&= 2.767mA
\end{aligned}
$$

$$
\begin{aligned}
r_e &= \frac{26mV}{I_E} \\
&= \frac{26mV}{2.767mA} \\
&= 9.39\Omega
\end{aligned}
$$

$$(\beta+1)r_e = (161)(9.39\Omega) = 1.51\ k\Omega$$

(b) Input Impedance

$$Z_i = R_{F_1}||(\beta+1)r_e = 120k\Omega||1.51\ k\Omega \approx 1.49k\Omega$$

The ac equivalent network is given in Fig. 2

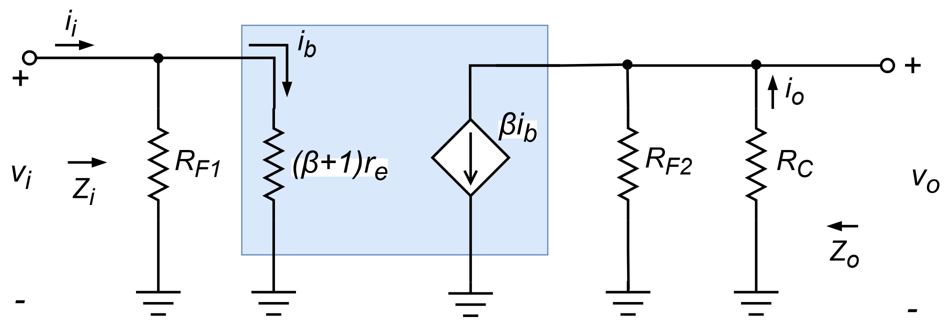


Figure 2

(c) Output Impedance

$$Z_o \approx R_C||R_{F_2} = 3k\Omega||68k\Omega = 2.87k\Omega$$

(d) Voltage gain

$$V_i = I_b(\beta+1)r_e$$

$$V_o = \beta I_b(R_{F_2}||R_C)$$

$$\begin{aligned} A_v &= -\frac{R_{F_2}||R_C}{r_e} * \frac{\beta}{\beta+1} \\ &= -\frac{68k\Omega||3k\Omega}{9.39\Omega} * \frac{160}{161} \\ &= -\frac{2.87k\Omega}{7.53\Omega} * 0.9929 \\ &= -305.64 * 0.9937 \\ &= -303.71 \end{aligned}$$

(e) Current gain

$$i_b = \frac{120}{121.51} i_i$$

$$i_o = (\beta i_b)\frac{68}{71}$$

$$i_o = \frac{160 * 120}{121.51} * \frac{68}{71} i_i$$

$$A_i = 151.33$$

2. A mild steel ring has a radius of 50 mm and a cross-sectional area of 400 mm^2. A current of 0.5 A flows in a coil wound uniformly around the ring and the flux produced is 0.1 mWb. If the relative permeability at this value of current is 200, find (a) the reluctance of the mild steel and (b) the number of turns on the coil.
Solution:

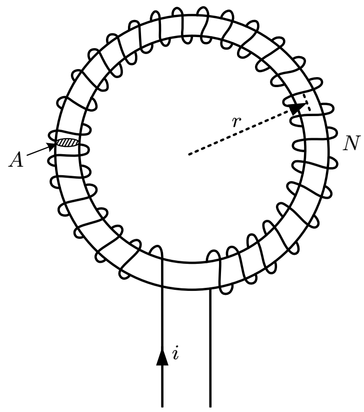


Figure 3

$$l = 2\pi r = 2 \times \pi \times 50 \times 10^{-3}\, m;\ A = 400 \times 10^{-6}\, m^2;$$
$$I = 0.5\, A;\ \phi = 0.1 \times 10^{-3}\, Wb;\ \mu_r = 200$$

Part a:

$$\text{Reluctance } R = \frac{l}{\mu_0 \mu_r A} = \frac{2 \times \pi \times 50 \times 10^{-3}}{(4\pi \times 10^{-7})(200)(400 \times 10^{-6})}$$
$$= 3.125 \times 10^6 / H$$

Part b:

$$R = \frac{\text{MMF}}{\Phi} \Rightarrow \text{MMF} = R\phi$$
$$\text{Again, } NI = R\phi$$

$$\text{Hence, } N = \frac{R\phi}{I} = \frac{3.125 \times 10^6 \times 0.1 \times 10^{-3}}{0.5} = 625 \text{ turns}$$

3. A section through a magnetic circuit of uniform cross-sectional area 2 cm^2 is shown in Figure 4(a). The cast steel core has a mean length of 25 cm. The air gap is 1 mm wide, and the coil has 5000 turns. The $B-H$ curve for cast steel is shown in Figure 4(b). Determine the current in the coil to produce a flux density of $0.80\, T$ in the air gap, assuming that all the flux passes through both parts of the magnetic circuit.
Solution:
For the cast steel core, when $B = 0.80\, T$, $H = 750\ At/m$
Reluctance of core,

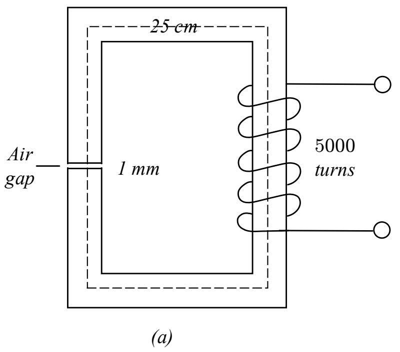
$$R_c = \frac{l_c}{\mu_0 \mu_r A_c} \text{ and } B = \mu_0 \mu_r H \Rightarrow \mu_r = \frac{B}{\mu_0 H}$$
$$\text{Thus, } R_c = \frac{l_c}{\mu_0 (\frac{B}{\mu_0 H}) A_c} = \frac{l_c H}{B A_c} = \frac{(25 \times 10^{-2})(750)}{(0.8)(2 \times 10^{-4})} = 1172000\, /H$$

For the air gap,

$$R_g = \frac{l_g}{\mu_0 \mu_r A_g} = \frac{l_g}{\mu_0 A_c} \quad (\text{since } \mu_r = 1 \text{ for air and } A_c = A_g)$$
$$= \frac{1 \times 10^{-3}}{(4\pi \times 10^{-7})(2 \times 10^{-4})} = 3979000\, /H$$

Total circuit reluctance,

$$R = R_c + R_g = (1172000 + 3979000) = 5151000 / H$$



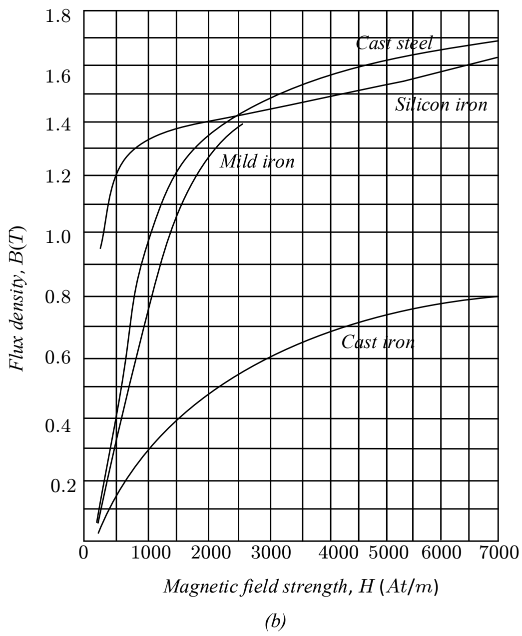


Figure 4 (a) Cross section of magnetic core, (b) B-H curves

Flux,

$$\phi = BA_c = 0.80 \times 2 \times 10^{-4} = 1.6 \times 10^{-4}\, Wb$$

Now,

$$R = \frac{\text{MMF}}{\phi} \Rightarrow \text{MMF} = R\phi \Rightarrow NI = R\phi$$

So,

$$I = \frac{R\phi}{N} = \frac{(5151000)(1.6 \times 10^{-4})}{5000} = \mathbf{0.165\ A}$$

4. The cross-sectional area of a transformer limb is 80 cm^2, and the volume of the transformer core is 5000 cm^3. The maximum value of the core flux is 10 mWb at a frequency of 50 *Hz*. Taking the Steinmetz constant as 1.7, the hysteresis loss is found to be 100 W. Determine the value of the hysteresis loss when the maximum core flux is 8 mWb and the frequency is 50 *Hz*.

Solution:
When the maximum core flux is 10 mWb:

$$\begin{aligned}
\text{Maximum flux density, } B_{m1} &= \frac{\phi_1}{A} = \frac{10 \times 10^{-3}}{80 \times 10^{-4}} = 1.25\, T \\
\text{Hysteresis loss, } P_{h1} &= k_h v f (B_{m1})^n \\
\Rightarrow 100 &= k_h (5000 \times 10^{-6})(50)(1.25)^{1.7} \\
\Rightarrow k_h &= \frac{100}{(5000 \times 10^{-6})(50)(1.25)^{1.7}} = 273.7
\end{aligned}$$

When the maximum core flux is 8 mWb:

$$\begin{aligned}
\text{Maximum flux density, } B_{m2} &= \frac{\phi_2}{A} = \frac{8 \times 10^{-3}}{80 \times 10^{-4}} = 1\, T \\
\text{Hysteresis loss, } P_{h2} &= k_h v f (B_{m2})^n \\
&= (273.7)(5000 \times 10^{-6})(50)(1)^{1.7} = \mathbf{68.4\ W}
\end{aligned}$$